

Course Syllabus

Course Description:

Data science refers broadly to using statistics and informatics techniques to gain insights from large datasets. Biomedical informatics refers to a range of disciplines that use computational approaches to analyze biomedical data to answer pre-specified questions as well as to discover novel hypotheses. In this course, we will use R and other freely available software to learn fundamental data science applied to a range of biomedical informatics topics, including those making use of health and omics data. After completing this course, students will:

- Be able to retrieve and clean data, perform exploratory analyses, build models to answer scientific questions, and present visually appealing results to accompany data analyses.
- Be familiar with various biomedical data types and resources related to them.
- Know how to create reproducible and easily shareable results with RStudio and GitHub.

Course Directors:

Brielin C Brown, PhD

Assistant Professor of Informatics, Genetics and Computer Science

5E 307

3600 Civic Center blvd

Philadelphia, PA 19104

brielin.brown@penmedicine.upenn.edu

[\(mailto:brielin.brown@penmedicine.upenn.edu\)](mailto:brielin.brown@penmedicine.upenn.edu)

Anurag Verma, PhD

Assistant Professor of Medicine and Informatics

A202 Richards Laboratory

3800 Hamilton Walk

Philadelphia, PA 19104

anurag.verma@pennmedicine.upenn.edu

[\(mailto:anurag.verma@pennmedicine.upenn.edu\)](mailto:anurag.verma@pennmedicine.upenn.edu)

Guest Lecturers:

Chloe Paris (chloe.paris@pennmedicine.upenn.edu

[\(mailto:chloe.paris@pennmedicine.upenn.edu\)](mailto:chloe.paris@pennmedicine.upenn.edu).)

John Holmes, PhD (jhholmes@pennmedicine.upenn.edu

[\(mailto:jhholmes@pennmedicine.upenn.edu\)](mailto:jhholmes@pennmedicine.upenn.edu).)

[\(mailto:dlmowery@pennmedicine.upenn.edu\)](mailto:dlmowery@pennmedicine.upenn.edu)

Joel Babdor, PhD (joel.babdor@pennmedicine.upenn.edu

[\(mailto:joel.babdor@pennmedicine.upenn.edu\)](mailto:joel.babdor@pennmedicine.upenn.edu).)

Danielle Mowery, PhD (dlmowery@pennmedicine.upenn.edu

[\(mailto:dlmowery@pennmedicine.upenn.edu\)](mailto:dlmowery@pennmedicine.upenn.edu).)

Walter Witschey, PhD (witschey@Pennmedicine.upenn.edu

[\(mailto:witschey@Pennmedicine.upenn.edu\)](mailto:witschey@Pennmedicine.upenn.edu).)

TAs:

Ananya Rajagopalan (Ananya.Rajagopalan@Pennmedicine.upenn.edu

[\(mailto:Ananya.Rajagopalan@Pennmedicine.upenn.edu\)](mailto:Ananya.Rajagopalan@Pennmedicine.upenn.edu).)

Office hours: Mondays 3-4pm in 3600 CCB room 03-304

Max Salvatore (Maxwell.Salvatore@Pennmedicine.upenn.edu

[\(mailto:Maxwell.Salvatore@Pennmedicine.upenn.edu\)](mailto:Maxwell.Salvatore@Pennmedicine.upenn.edu).)

Office hours: Wednesdays 11am-12pm in KRB room 1301

Format:

The course will consist of weekly in-person meetings on Thurs 3:30-6:30pm in Richards B102AB. For those with circumstances that preclude in person attendance, lectures may be recorded via Zoom.

Practicum materials to work through computational exercises will be available weekly. Six assignments will be due throughout the semester. A final project requiring a substantial amount of work and creativity will be due at the end of the semester. This project is in lieu of a final exam. Students will be encouraged to work independently and seek help as needed. Canvas will contain the latest course announcements.

Expectations:

You are expected to attend all the course sessions, participate in class discussions, and complete required exercises and the class project. This course requires use of a computer, which you must have to fully participate in lectures and online activities. You must be familiar with this computer and be able to install programs on it.

Grading: The course is graded on a letter grade basis, according to the following proportions:

40% assignments (6 total)

50% biomedical data science project

10% participation (attending synchronous sessions, providing feedback to peers)

Assignments:

Due dates: 9/2/26, 9/16/26, 9/30/26, 10/21/26, 11/4/26, 11/18/26.

Biomedical Data Science Project:

The final project will answer a question selected by each student using biomedical data and some of the tools presented during the course. After students choose the topic to address on their own, each will identify two faculty, staff and/or, postdocs to get feedback and help define a specific novel and interdisciplinary question. Although use of publicly available data for the project is encouraged, students may use an appropriate private data source. Students will work on these projects throughout the semester with final project reports due on 12/18/26. Grading will be based on three project components: (1) an html document derived from a Quarto Document that describes the question,

source of data, analysis, and results, (2) a GitHub repository that contains an organized project, and (3) a short presentation describing the work to classmates.

Textbooks:  <http://themlbook.com/wiki/doku.php>

There are many online resources to learn R, including topics beyond what we will cover in class. The following textbook is suggested but not required for those students who prefer to have a printed reference:

Wickham H, Çetinkaya-Rundel M, and Grolemund G, “R for Data Science (2e)” O’Reilly (2023)

Available online for free: <https://r4ds.hadley.nz/>  <https://r4ds.hadley.nz/>

For those seeking to a textbook that covers biostatistics in greater depth, we recommend the following one:

Rosner B, “Fundamentals of Biostatistics” Brooks/Cole, Cengage Learning. (2016)

For those seeking to a textbook that covers machine learning well and provides a starting point for further learning, we recommend the following one:

Burkov, A, “The Hundred-Page Machine Learning Book” (2019)

Available online for free: <https://themlbook.com/wiki/doku.php>  <https://themlbook.com/wiki/doku.php%C2%A0>

Prerequisites:

Familiarity with basic statistical (e.g., EPID 526/7 or other first-year graduate level stats course) concepts is expected, as this course will not cover basic concepts in depth. A background in biology and computing would be helpful, but no formal requirements will be enforced.

Academic Honesty:

All work submitted for credit is expected to be a student’s own work. In the preparation of all papers and other written work, students should always take great care to distinguish their own ideas and knowledge from information derived from other sources. The term “sources” includes not only published primary and secondary material, but also information and opinions gained directly from other

people. Should you use a tool based on large language models, it is your responsibility to verify the accuracy of contents provided and to ensure that you properly attribute any original sources of content identified. The responsibility for learning the proper forms of citation lies with each student. Students must acknowledge any collaboration and its extent in all submitted work. Students are expected to follow Penn's standards of academic integrity as found in:

<https://catalog.upenn.edu/pennbook/code-of-academic-integrity>

Policy on the use of Large Language Models:

Students are free to use Large Language Models (commonly called "AI") as a learning aid. This includes discussing class topics, reinforcing class ideas, or brainstorming ideas for both the homeworks and project assignments.

Because this is an introductory course, a core goal is to learn programming and data analysis fundamentals without reliance on AI. Copying course assignments into LLMs or submitting LLM generated code or text for assignments is forbidden, and anyone caught doing so will receive a 0 for the corresponding assignment.

Students with Disabilities:

The University of Pennsylvania provides reasonable accommodations to students with disabilities who have self-identified and been approved by the office of Student Disabilities Services (SDS). Please make an appointment to meet with me as soon as possible in order to discuss your accommodations and your needs. If you have not yet contacted SDS, and would like to request accommodations or have questions, please visit the SDS website at

<https://weingartencenter.universitylife.upenn.edu/academic-accommodations/>

(<https://weingartencenter.universitylife.upenn.edu/academic-accommodations/>) to find out more. The SDS office is located in the Weingarten Center at Stouffer Commons, 3702 Spruce Street, Suite 300. All SDS services are free and confidential.

Course calendar:

Week 1: August 27th - Intro / Reproducible research [Practicum 1/2]

Assignment 1 Due September 2nd.

Week 2: September 3rd - R I / II [Practicum 3/4]

Week 3: September 10th - Information retrieval / Cleaning [Practicum 5/6]

Assignment 2 Due September 16th.

Week 4: September 17th - Data visualization / Exploratory data analysis [Practicum 7/8]

Week 5: September 24th - Regression / Model evaluation [Practicum 9/10]

Assignment 3 Due September 30th

Week 6: October 1st - Fall Break NO CLASS

Week 7: October 8th - ML overview / Supervised ML I [Practicum 11]

Week 8: October 15th - Supervised ML II / Unsupervised ML [Practicum 12]

Assignment 4 Due October 21st

Week 9: October 22nd - Womens Health / Epidemiology and EHRs [Practicum 13]

Week 10: October 29th - Geospatial data / Environmental data [Practicum 14/15]

Assignment 5 Due November 4th

Week 11: November 5th - Omics / Immunomics [Practicum 16/17]

Week 12: November 12th - Secondary Health Data / Biobanks and Genetic Variation

Assignment 6 Due November 18th

Week 13: November 19th - Medical text / Generative AI

Week 14: November 26th - Thanksgiving break NO CLASS

Week 15: December 3rd - Imaging data / AI for biomedical informatics

Week 16: December 10th - Project presentations

Week 17: December 17th - Project presentations